

Rapport: Monostable Multivibrator Using NAND

1 Objective

The objective of this work is to simulate and analyze the behavior of the electronic circuit shown in the provided schematic. The studied circuit operates as a **monostable multivibrator**, also known as a **one-shot circuit**.

A monostable multivibrator is characterized by having:

- One stable state
- One unstable (temporary) state

When triggered by an input pulse, the circuit leaves its stable state for a fixed duration and then automatically returns to the stable state.

2 Main Components

The circuit consists of the following primary elements:

- **Two logic gates:** Represented by the yellow blocks in the schematic
- **One capacitor (C):** Used for timing
- **One resistor (R):** Used for timing
- **Common ground reference**

2.1 Voltage Observation Points

The simulation tracks four specific nodes:

1. Input logic signal to the first logic gate
2. Output of the first logic gate and input to the capacitor
3. Output of the second logic gate
4. Output voltage across the RC network

3 Circuit Operation and Simulation

The simulation was performed using **Falstad**, an online simulator that allows for real-time visualization of voltages and currents.

3.1 Simulation Setup

- The circuit was constructed by placing logic gates, resistors, capacitors, and voltage sources according to the schematic
- **Digital logic gates** generate binary signals (HIGH/LOW), while **analog components** (R and C) determine the time-dependent behavior
- The input is driven by a pulse-generation circuit that produces a periodic digital signal without an external clock source

3.2 Role of the RC Network

The duration of the pulse is governed by the resistor and capacitor:

- A **larger resistance** increases the pulse duration
- **Smaller values** result in faster oscillations or shorter pulses

4 Observed Outputs

The simulation provides visual data for the behavior of each node:

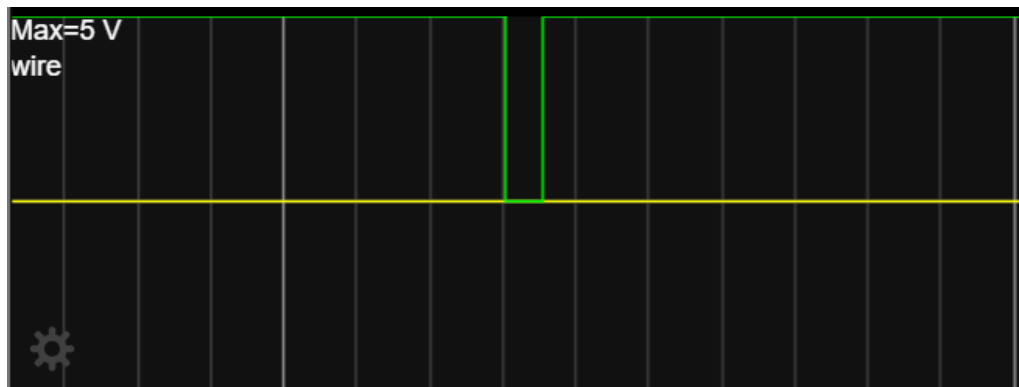


Figure 1: Voltage waveform at observation point 1



Figure 2: Voltage waveform at observation point 2

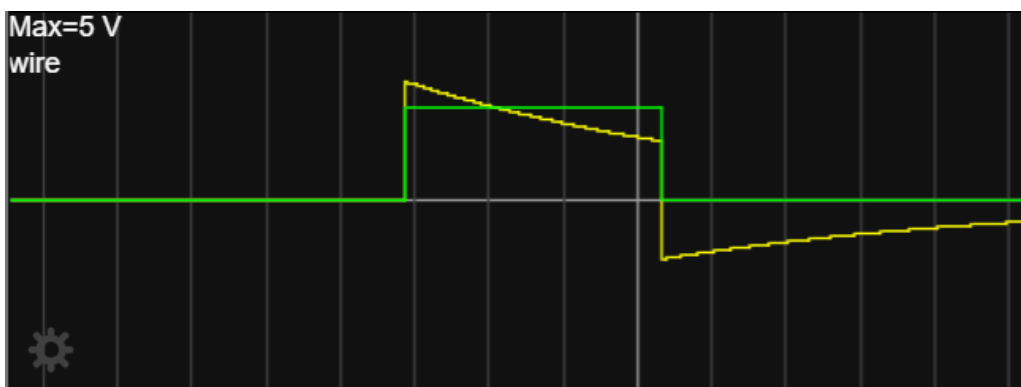


Figure 3: Voltage waveform at observation point 3

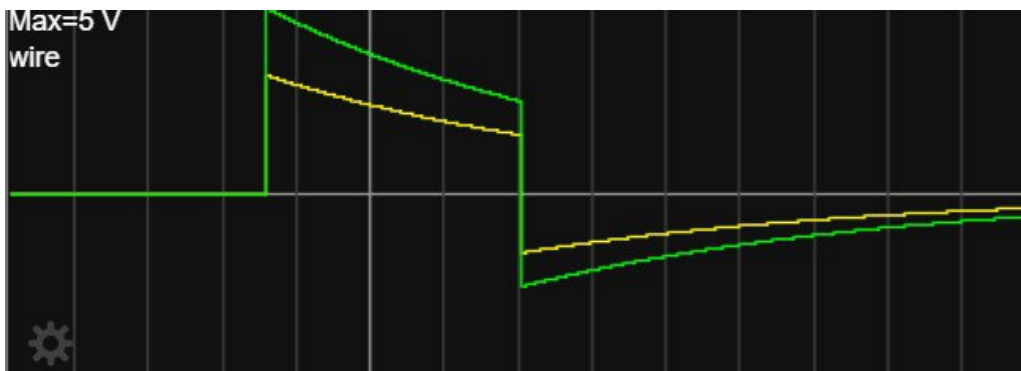


Figure 4: Voltage waveform at observation point 4

5 Conclusion

The simulation confirms that the circuit behaves as a **monostable multivibrator**. Upon receiving a trigger, the output generates a pulse of a fixed duration defined by the RC network and then returns to its stable state. This observed behavior matches the theoretical operation of monostable circuits.